Prischa Bajeli

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Education

MIT World Peace University — Pune, India
B.Tech - Computer Science & Engineering (AI & Data Science) — 2023 – Present

- **CGPA:** 8.20 | **2x published researcher** (IEEE ETCOM 2025, Springer LNNS ICTCS 2025)
- **Relevant courses:** Data Structures & Algorithms, DBMS, Data Engineering, Machine Learning, Deep Learning, AI & Expert Systems, Cognitive Computing & NLP, Operating Systems, Theory of Computation

Indian School Muscat — Muscat, Oman
CBSE – Class 12: 84% Class 10: 94% — 2008 – 2023

Skills

Languages: Python, C, C++, SQL, TypeScript, JavaScript

ML & Deep Learning: PyTorch, Keras, TensorFlow, scikit-learn, LSTM, DenseNet121, Computer Vision, NLP

AI & LLM Engineering: Anthropic Claude API, LangGraph, RAG, ChromaDB, sentence-transformers, agentic tool-calling, multimodal/vision

Data & Analysis: pandas, NumPy, Matplotlib, seaborn, EDA, feature engineering, Excel, IBM Cognos

Web & Full-Stack: Next.js, React Native, Expo, Express.js, FastAPI, PostgreSQL, Prisma, Drizzle ORM, Clerk

MLOps & Tools: Docker, GitHub Actions (CI/CD), pytest, Git, Jupyter, Raspberry Pi

Soft Skills: Technical writing, Research & experimentation, Leadership, Communication

Projects

Explainable AI for Chest X-Ray Diagnosis (Deep Learning + XAI) — Github
Python | PyTorch | DenseNet121 | FastAPI | Grad-CAM | NumPy | Matplotlib

- Designed a multi-label DenseNet121 classifier for the NIH ChestX-ray14 dataset (112k images, 14 pathology labels) using patient-level stratification across splits to avoid data leakage; obtained 0.83 macro-AUC, matching benchmark results without pre-training tricks.
- Calibrated label-specific thresholds via ROC curves on the validation set to optimize F1 for rare pathology detection while minimizing false negatives in critical cases.
- Deployed the model via a FastAPI REST endpoint with a Grad-CAM web UI, letting non-technical users visualize predictions alongside per-label probability distributions.

Siora2k5 — Agentic AI Research Assistant — Github
Python | Claude API | LangGraph | FastAPI | Next.js | ChromaDB | Tavily

- Built an agentic AI assistant that autonomously selects and chains tools (web search, calculation, document RAG, image vision) to answer multi-step questions, implemented both from scratch and with LangGraph to demonstrate agentic loops and the production framework pattern.
- Engineered a RAG pipeline (ChromaDB + sentence-transformers) with per-message multimodal vision over a FastAPI backend and custom Next.js chat UI, plus an evaluation harness measuring tool-selection accuracy (100% on the test set).

MedGuide — Clinical Guideline RAG Assistant — Github
Python | Claude API | ChromaDB | sentence-transformers | PyMuPDF | FastAPI

- Built a retrieval-augmented generation system answering medical questions grounded in real WHO clinical guidelines, returning plain-language answers with inline citations and correctly refusing when the corpus lacks the information.
- Improved retrieval quality by switching from per-page to cross-document chunking with a minimum-length filter, removing ∼500 noise fragments and substantively improving answers with no model or prompt change.

- Built an evaluation harness scoring retrieval accuracy, citation grounding, and refusal accuracy (12/12 on the test set), treating evaluation as infrastructure.

JobAI — AI-Powered Career Copilot Github | Live App

TypeScript | React Native | Expo | Express.js | PostgreSQL | Drizzle ORM | Anthropic Claude API | TanStack React Query

- Developed a cross-platform mobile career assistant in React Native/Expo, integrating the Anthropic Claude API to tailor resumes, cover letters, and cold-outreach messaging for users.
- Created a type-safe backend with Express.js, PostgreSQL, and Drizzle ORM, with a job-application tracker that persists application status and contact history across sessions, and TanStack React Query managing client-side caching and sync with the LLM-backed API.

Credit Risk Analysis — Predicting Loan Default Github

Python | pandas | NumPy | scikit-learn | Matplotlib | seaborn

- Ran an end-to-end data science analysis on 150K borrowers (Give Me Some Credit) to identify default drivers, with EDA, median imputation, and class-balanced modeling on a highly imbalanced target (6.7% default rate).
- Compared Logistic Regression vs. Random Forest on ROC-AUC, precision, and recall (Random Forest ROC-AUC 0.84), framing the recall-vs-precision trade-off around the lender's cost of a missed defaulter vs. a wrongly rejected applicant.
- Surfaced credit utilization as the top non-linear driver of default via Random Forest feature importance despite near-zero linear correlation, demonstrating where correlation alone misleads.

Heart Disease MLOps Pipeline Github

Python | scikit-learn | FastAPI | Docker | GitHub Actions | pytest

- Built an end-to-end ML pipeline that trains a Random Forest on the UCI Heart Disease dataset and serves predictions through a FastAPI REST API with a typed request schema.
- Containerized the service with Docker and automated testing and image builds on every push via a GitHub Actions CI workflow, demonstrating a production-style train-to-deploy loop.

Multivariate Time-Series Energy Consumption Prediction Github | Project Blog

Python | Keras | TensorFlow | LSTM | scikit-learn | pandas | NumPy

- Developed an RFE-LSTM forecasting pipeline on 35K hourly energy samples; an ablation study over feature and window counts cut MAE/RMSE by up to 7% vs. the no-lag baseline, identifying short lag features (lag 1–3) as the dominant signals. *Published at ICTCS 2025 (see Publications).*

IoT-Enabled Face Recognition Attendance System Github | Project Blog

Python | OpenCV | LBPH | Raspberry Pi 4 | GPIO | SQLite

- Built an end-to-end LBPH face recognition pipeline on Raspberry Pi 4 achieving 95%+ accuracy on 50+ student profiles, reducing per-session logging from ~5 minutes to zero manual input. *Published at IEEE ETCOM 2025 (see Publications).*

Archive 2k5 — Full-Stack Thrift E-Commerce Platform Github

Next.js | Prisma | PostgreSQL/SQLite | Clerk | Razorpay

- Built a full-stack e-commerce thrift store with Next.js, Prisma, and Clerk authentication, featuring an item-reservation flow that locks inventory at checkout to prevent double-purchase.
- Implemented a custom UPI QR manual-verification checkout with an admin order-confirmation workflow, plus a dormant Razorpay integration switchable via a single environment flag with no code changes.

Publications

IoT-Enabled Face Recognition Attendance System Link

Prischa Bajeli et al., IEEE ETCOM 2025

- Presents an edge-deployed attendance system that runs LBPH-based facial recognition entirely on a Raspberry Pi 4, achieving 95%+ recognition accuracy across 50+ student profiles while removing all manual logging from the attendance workflow.

Multivariate Time-Series Energy Consumption Prediction Link

Prischa Bajeli et al., Springer LNNS, ICTCS 2025, Jaipur, India

- Proposes an RFE-LSTM forecasting pipeline trained on 35K hourly energy samples, using recursive feature elimination and an ablation study over feature and window counts to cut MAE/RMSE by up to 7%, identifying short lag features (lag 1–3) as the dominant predictive signals.

Certifications

AI Fundamentals – IBM SkillsBuild • 2024

Data Visualizations and Building Dashboards with Excel and Cognos • 2024

NPTEL – Introduction to Large Language Models • 2026